

Loan Risk Analytics – EDA Assignment

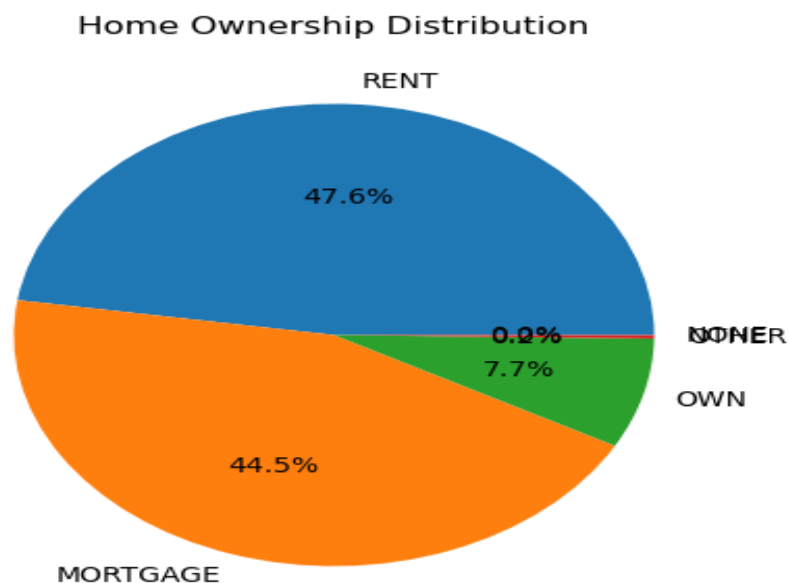
FINAL INSIGHTS

Problem Statement: For the given dataset consisting of loan details of various borrowers from 2007 to 2011, the objective is to find risky loan applicants. That is, the ones who are most likely to default the loan. Identification of such applicants, or in other words, the driving factors behind loan default using EDA is the aim of this assignment.

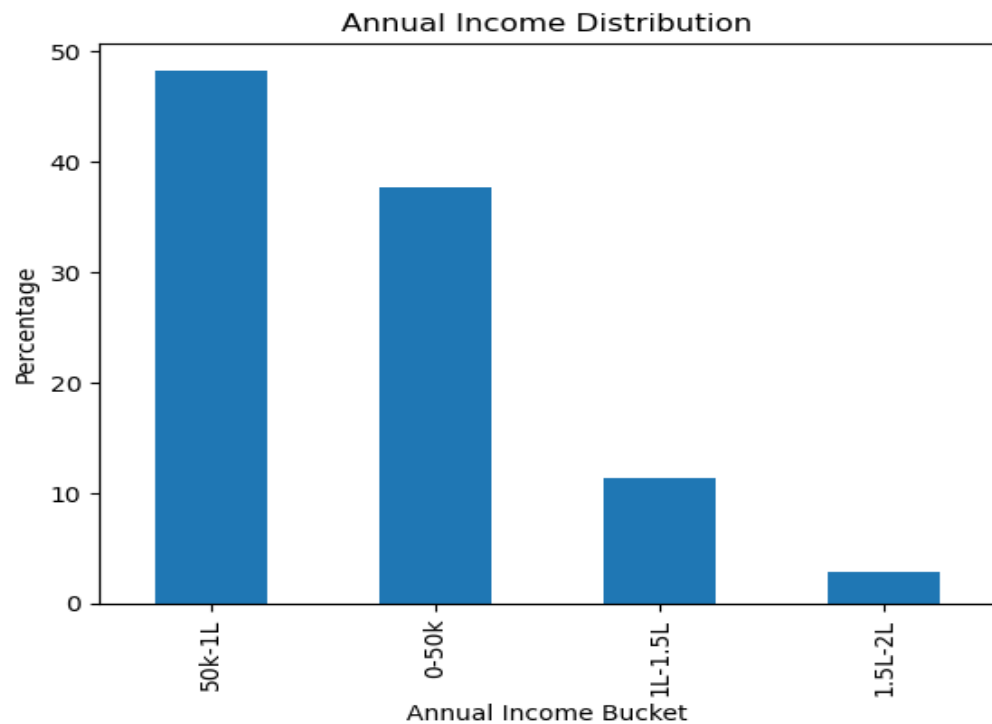
The company will utilize this knowledge for its portfolio and risk assessment in the future, so that its credit loss can be minimised.

Analysis Approach:

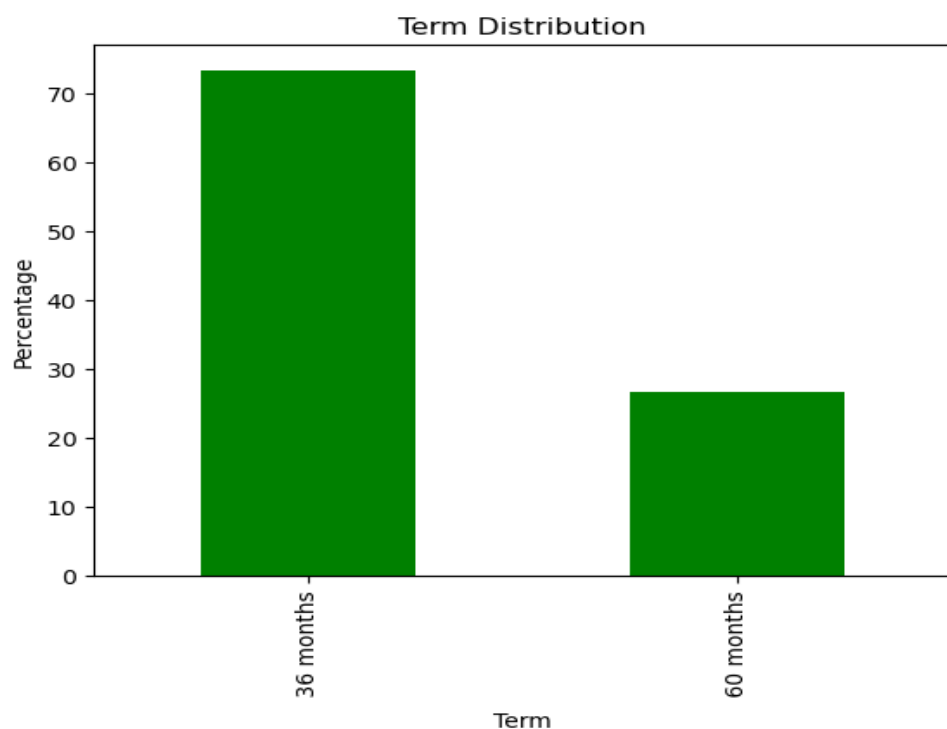
1. After importing the necessary libraries, I have read the given 'loan' dataset into a *pandas* dataframe. It consists of 111 columns and 39717 rows, of which many are all-null columns.
2. **Identifying target variable** – 'loan_status' is the target variable here. The aim is to study how various factors impact this variable.
3. **Handling missing values** – a) The first step I did was to identify and delete all completely null columns since they are of no use. There are also many unnecessary columns that do not add any value in analysis, which have been deleted. Now the count of columns has come down to 43.
b) Next, for better convenience during statistical analysis, the null values in other numerical columns have been replaced with zeroes.
4. **Univariate Analysis & Outlier Handling** – Insights discovered from these analyses are as below.
 - a) Most borrowers from our dataset have a rented home (47.6%), followed by those who currently have a home mortgage (44.5%).



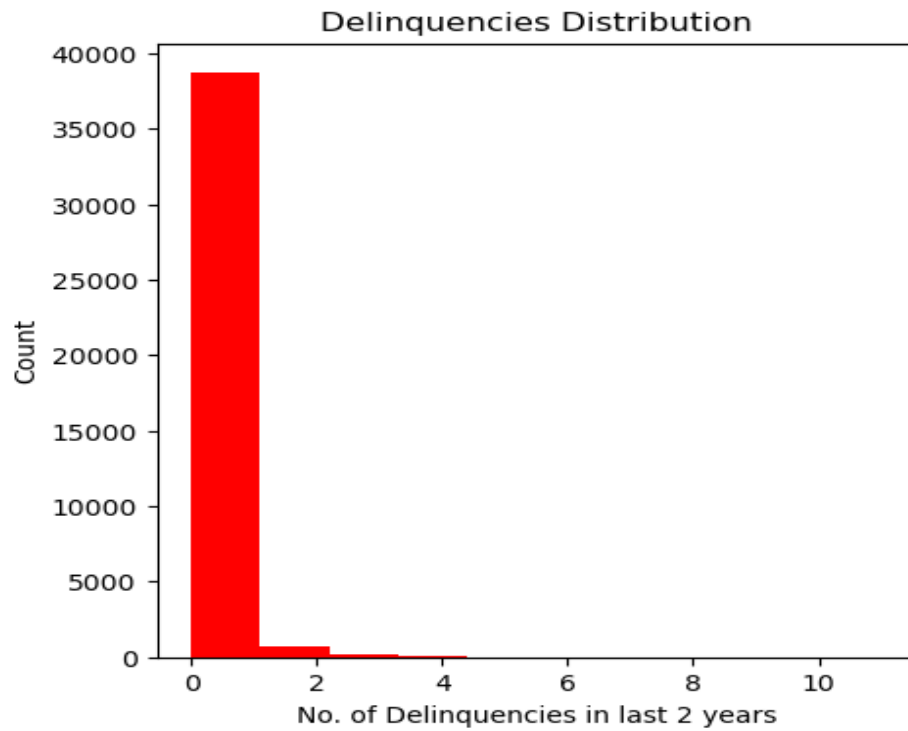
b) Most borrowers fall under the 50k-1lakh annual income bucket, followed by the 0-50k bucket.



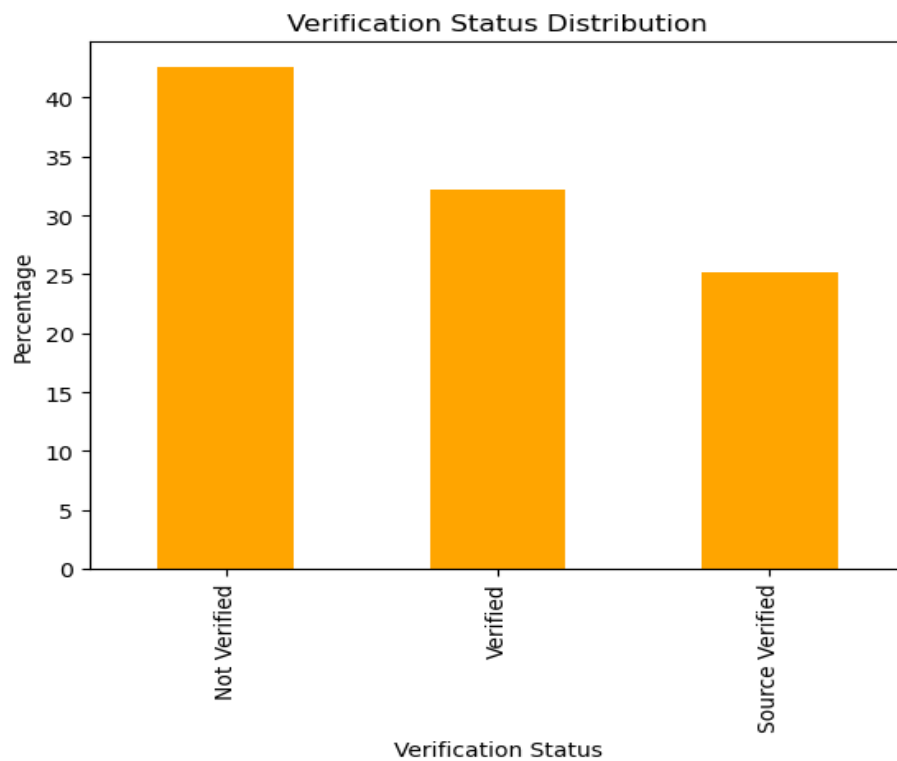
c) Almost 75% of the loans being given are for a period of 36 months. The rest 25% are for 60 months.



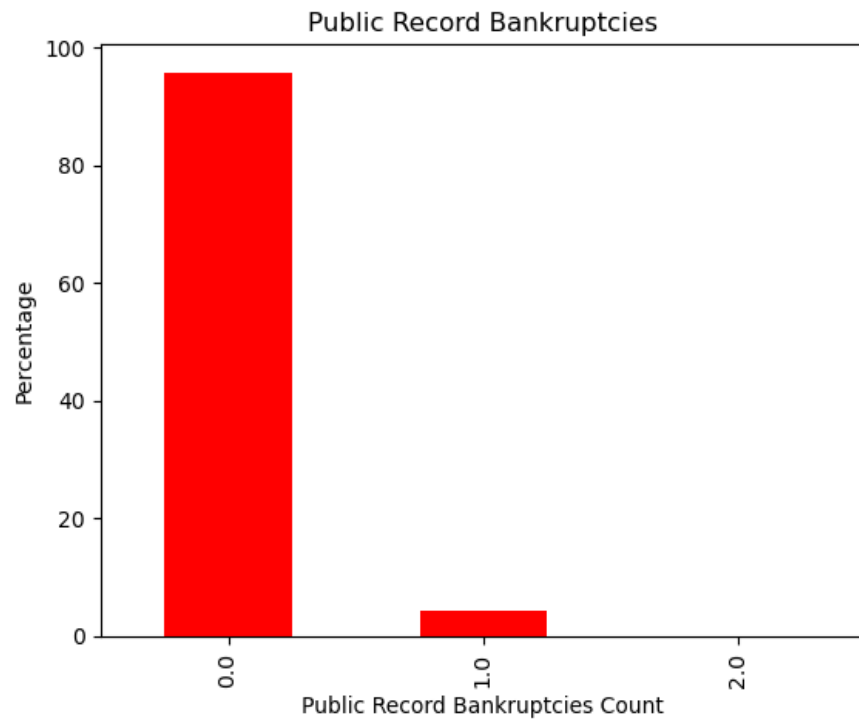
d) Most borrowers (89.2%) do not have any past delinquencies in the last 2 years, which is a good thing. The remaining 10.8% of borrowers can be considered cautiously, as they are risky.



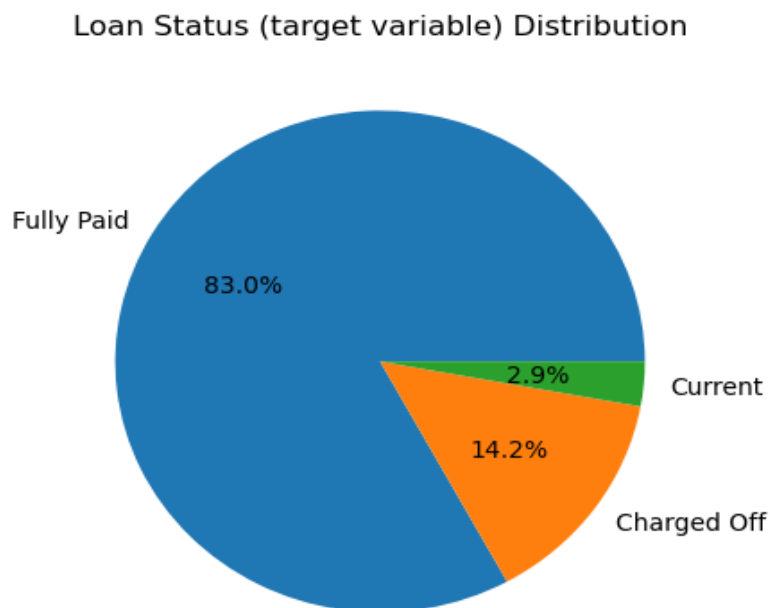
e) We have more than 40% of borrowers whose income source is not verified.



f) Over 95% of borrowers do not have any public recorded bankruptcies. The other 5% can be considered risky customers.

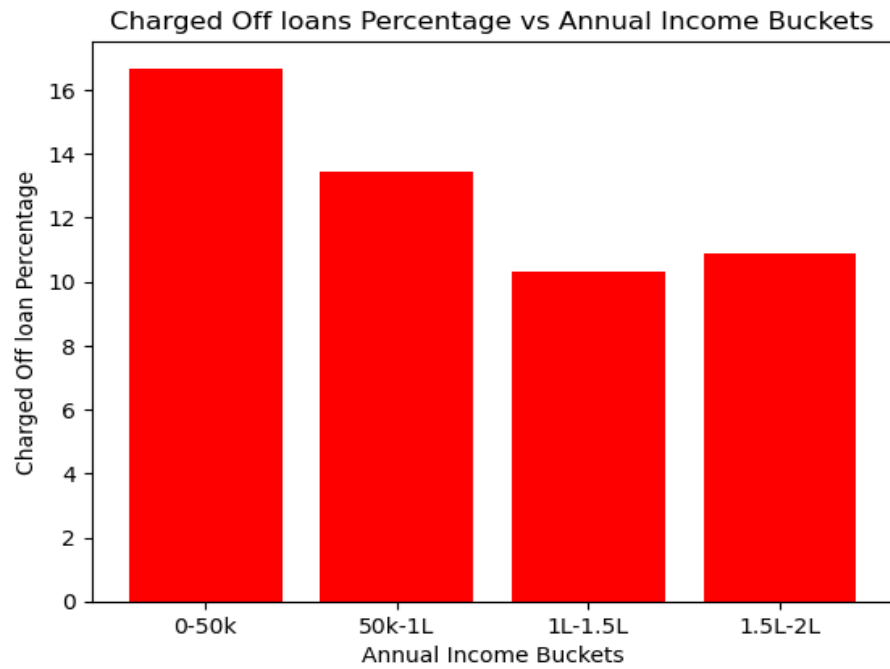


g) As much as 14.2% loans are charged off, which is a loss to the company. This needs to be minimized.

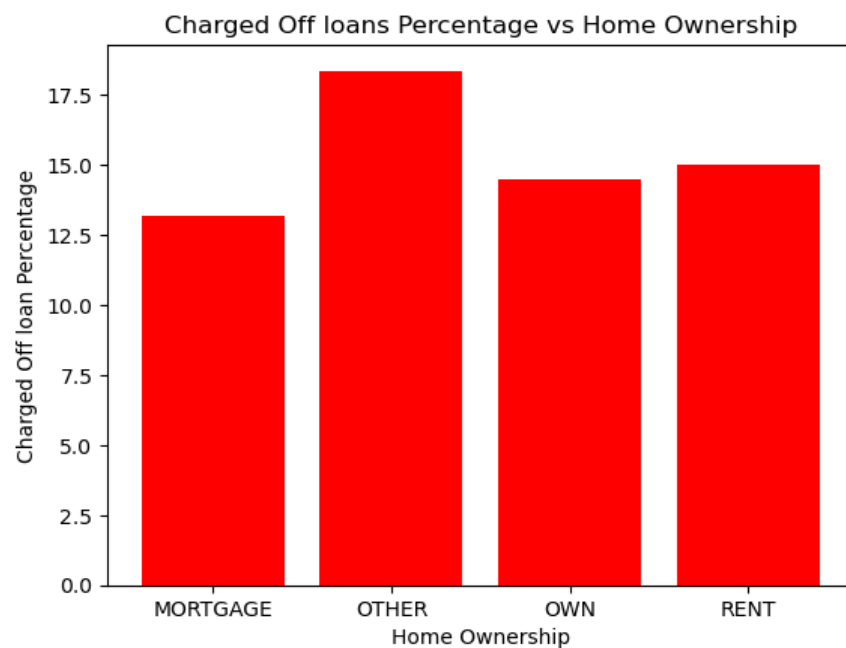


5. Bivariate Analysis – Insights discovered are as follows.

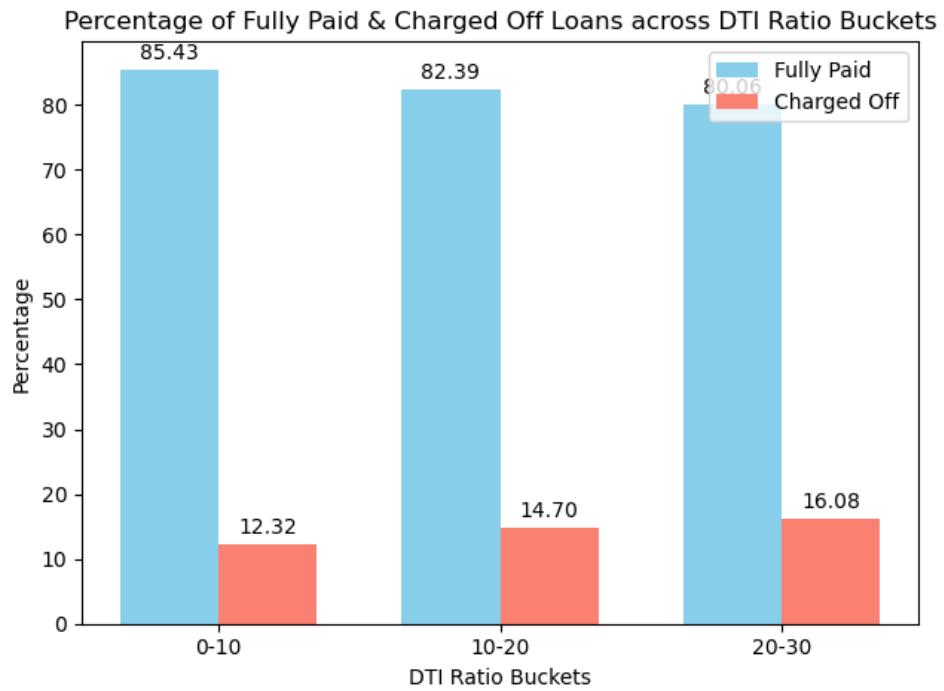
- a) **Borrowers belonging to lower income range have higher chances of defaulting loans, and can be considered risky.** This is because 'Charged Off' loan percentage is highest for borrowers having 0-50k annual income range.



- b) Borrowers with rented house/others/none are riskier than those with an own house or a mortgage. Hence, the latter should be preferred.



- c) Borrowers with low DTI (debt-to-income) ratio are more preferable, as charged off loan percentage increases with increase in DTI ratio.



- d) As much as possible, offer shorter term loans i.e., for a term of 36 months, rather than for 60 months. Loans with longer terms have a **SIGNIFICANTLY HIGHER** charge off rate than those with **shorter terms**! This is a very important insight to consider.

